Materials Design Analysis Reporting (MDAR) Checklist for Authors

The MDAR framework establishes a minimum set of requirements in transparent reporting applicable to studies in the life sciences (see Statement of Task: doi:10.31222/osf.oio/9sm4x.). The MDAR checklist is a tool for authors, editors and others seeking to adopt the MDAR framework for transparent reporting in manuscripts and other outputs. Please refer to the MDAR Elaboration Document for additional context for the MDAR framework.

Materials

Antibodies	Yes (indicate where provided: page no/section/legend)	n/a
For commercial reagents, provide supplier name, catalogue number and RRID, if available	No antibodes detected. Please add identifiers for all resources where possible	

Cell Materials	Yes (indicate where provided: page no/section/legend)	n/a
Cell lines: Provide species information, strain. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No cell lines detected Please add identifiers for all resources where possible	
Primary cultures: Provide species, strain, sex of origin, genetic modification status.	Not currently checked by SciScore	

Experimental Animals	Yes (indicate where provided: page no/section/legend)	n/a
Laboratory animals: Provide species, strain, sex, age, genetic modification status. Provide accession number in repository OR supplier name, catalog number, clone number, OR RRID	No organisms detected Please add identifiers for all resources where possible	
Animal observed in or captured from the field: Provide species, sex and age where possible	Not currently checked by SciScore	
Model organisms: Provide Accession number in repository (where relevant) OR RRID	See laboratory animals section for information.	

Plants and microbes	Yes (indicate where provided: page no/section/legend)	n/a
Plants: provide species and strain, unique accession number if available, and source (including location for collected wild specimens)	Not currently checked by SciScore	
Microbes: provide species and strain, unique accession number if available, and source	Not currently checked by SciScore	

Human research participants	Yes (indicate where provided: page no/section/legend)	n/a
Identify authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Provide statement confirming informed consent obtained from study participants.	Not detected.	
Report on age and sex for all study participants.	**Age:**Quantitative Reverse Transcription Polymerase Chain Reaction (qRT-PCR) A total of 8 dbdb mice and 8 heterozygous mice at the age of 8 weeks were used. **Sex:**Male dbdb mice were used as animal model of T2D , and the heterozygous mice were used as control .	

Design

Study protocol	Yes (indicate where provided: page no/section/legend)	n/a
For clinical trials, provide the trial registration number OR cite DOI in manuscript.	Not detected.	

Laboratory protocol	Yes (indicate where provided: page no/section/legend)	n/a
Provide DOI or other citation details if detailed step-by-step protocols are available.	Not detected.	

Experimental study design (statistics details)	Yes (indicate where provided: page no/section/legend)	n/a
State whether and how the following have been done, or if they were not carried out		
Sample size determination	not detected.	
Randomization	not detected.	
Blinding	not detected.	
inclusion/exclusion criteria	not detected.	

Sample definition and in-laboratory replication	Yes (indicate where provided: page no/section/legend)	n/a
State number of times the experiment was replicated in laboratory	Not detected.	
Define whether data describe technical or biological replicates	Not detected.	

Ethics	Yes (indicate where provided: page no/section/legend)	n/a
Studies involving human participants: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	Not detected.	
Studies involving experimental animals: State details of authority granting ethics approval (IRB or equivalent committee(s), provide reference number for approval.	The animal studies were approved by the Animal Care and Ethics Committee of West China Hospital of Stomatology, Sichuan University.	
Studies involving specimen and field samples: State if relevant permits obtained, provide details of authority approving study; if none were required, explain why.	Not detected.	

Dual Use Research of Concern (DURC)	Yes (indicate where provided: page no/section/legend)	n/a
If study is subject to dual use research of concern, state the authority granting approval and reference number for the regulatory approval	Not currently checked by SciScore	

Analysis

Attrition	Yes (indicate where provided: page no/section/legend)	n/a
State if sample or data point from the analysis is excluded, and whether the criteria for exclusion were determined and specified in advance.	not detected.	

Statistics	Yes (indicate where provided: page no/section/legend)	n/a
Describe statistical tests used and justify choice of tests.	Considering that multiple tests are used among the three datasets , we use the Bonferroni adjustment to set a more stringent threshold , p < 0.0167 (0.053) was set as a threshold for defining differential genes .	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
State whether newly created datasets are available, including protocols for access or restriction on access.	6 The data is publicly available on the DIAGRAM website (http:diagram-consortium.orgindex.html).	
If data are publicly available, provide accession number in repository or DOI or URL.	Table 1 The 20 Most Important m6A-SNPs Related to T2D Variant CHR Position P . value Gene Confidence_Level eQTL Hits DEG Methylation Type Allele Frequency rs1801206 4 6302707 4.90E-39 WFS1 Prediction: (Low) 8 hits No m6A 5.76E-01 rs116234738 5 102537357 1.40E-27 PPIP5K2 Prediction: (Low) 0 hit Yes m6A 3.07E-02 rs5213 11 17408404 Medium) 16 hits Yes m6A 7.18E-01 rs750625 8 41525914 ANK1 Prediction: (Low) 3 hits No m6A 6.57E-06 rs10832778 11 17394073 8.50E-23 NCR3LG1 Prediction: (Low) 26 hits No m6A 7.09E-01 rs9379084 6 7231843 RREB1 MeRIP-Seq: (Medium) 1 hit Yes m6A 9.19E-02 rs12590 10 12292348 5.50E-17 CDC123 Prediction: (Low) 2 hits No m6A 1.50E-01 rs3743481 15 77907145 1.60E-16 LINGO1 MeRIP-Seq: (Medium) 1 hit No m6A 4.42E-01 rs2641348 1 120437884 5.60E-16 ADAM30 Prediction: (Low) 6 hits No m6A 1.69E-01 rs6685892 1 120458004 6.20E-16 NOTCH2 Prediction: (Low) 5 hits No m6A 1.68E-01 rs2793823 1 120437718 7.30E-15 ADAM30 Prediction: (Low) 3 hits No m6A 2.70E-01 rs11553326 4 724884 5.50E-12 PCGF3 Prediction: (Low) 7 hits No m6A 3.21E-02 rs2308891 6 32610009 HLA-DQA1 Prediction: (Low) 293 hits No m6A 2.44E-02 rs3760471 17 4069735 2.80E-11 ANKFY1 Prediction: (Low) 5 hits Yes m6A 1.38E-04 rs1058129 15 77905661 4.30E-11 LINGO1 Prediction: (Low) 5 hits No m6A 7.06E-01 rs7250850 19 47585517 9.70E-11 ZC3H4 Prediction: (Low) 1 hit No m6A 5.23E-01 rs1046080 6 31595882 1.30E-10 PRRC2A miCLIP: (High) 38 hits No m6A 7.43E-01 rs1033500 6 32307382 3.00E-10 C6orf10 Prediction: (Low) 21 hits No m6A 3.37E-01 rs2227956 6 31778272 3.10E-10 HSPA1L Prediction: (Low) 112 hits No m6A 8.77E-01 rs4993986 6 32627652 6.10E-09 HLA-DQB1 miCLIP: (High) 1 hit Yes m6A 3.69E-01 Prediction of m6A Modification Near m6A-SNPs Among the T2D-associated m6A-SNPs identified above , we predicted the potential m6A modification site on the transcript by inputting the reference	

Data availability	Yes (indicate where provided: page no/section/legend)	n/a
	and altered sequence to an online m6A modification prediction tool , SRAMP (http:www.cuilab.cnsramp). Three datasets GSE23343 , GSE38642 and GSE78721 containing standardized T2D-associated gene expression signals were retrieved and downloaded from GEO database (http:www.ncbi.nlm.nih.govgeo) . Also , to prove that these gene expression differences were mediated by m6A-SNPs instead of the T2D phenotype , we downloaded the transcriptome data of tissues that are not closely related to the T2D phenotype as control sets , including enteroendocrine cells (GSE132831) and peripheral blood monocytes (GSE156061) . For single-cell RNA sequencing (scRNA-seq) data analysis , we downloaded the islet scRNA-seq data of high-fat diet-induced diabetic mice and healthy controls in GSE162512 , and used the Seurat package to load the hdf5 format data for subsequent process.	
If publicly available data are reused, provide accession number in repository or DOI or URL, where possible.	Not detected.	

Code availability	Yes (indicate where provided: page no/section/legend)	n/a
For all newly generated code and software essential for replicating the main findings of the study:		
State whether the code or software is available.	Not detected.	
If code is publicly available, provide accession number in repository, or DOI or URL.	Not detected.	

Analysis

Adherence to community standards	Yes (indicate where provided: page no/section/legend)	n/a
MDAR framework recommends adoption of discipline-specific guidelines, established and endorsed through community initiatives. Journals have their own policy about requiring specific guidelines and recommendations to complement MDAR.		
State if relevant guidelines (eg., ICMJE, MIBBI, ARRIVE) have been followed, and whether a checklist (eg., CONSORT, PRISMA, ARRIVE) is provided with the manuscript.	Not currently checked by SciScore	